

BOJNORD, Iran, Islamic Republic of

WMO: 407230

Lat: 37.493N

Lon: 57.308E

Elev: 3543

StdP: 12.91

Time Zone: 3.50 (IRN)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.9	17.4	4.6	7.9	18.6	9.8	10.3	25.3	24.8	42.5	21.7	47.2	1.2	270	0.487

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.7	96.9	64.3	93.6	63.3	91.4	62.9	70.8	85.7	69.2	83.9	67.6	82.0	7.9	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.4	111.1	77.7	64.5	103.9	76.4	62.8	97.5	75.4	37.1	85.3	35.6	83.9	34.2	81.9	79.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
22.2	18.6	16.2	DB	8.3	101.6	6.4	2.7	3.7	103.6	0.0	105.2	-3.6	106.7	-8.2	108.7
			WB	7.2	73.5	7.2	2.1	2.0	75.0	-2.2	76.2	-6.2	77.4	-11.4	78.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	56.8	34.3	37.7	47.1	55.7	66.1	74.1	78.6	76.1	69.3	58.1	45.2	37.9
	DBStd	16.74	8.32	8.57	7.93	7.75	5.07	5.14	4.21	4.16	5.00	6.63	7.23	6.69
	HDD50	1589	487	350	149	35	0	0	0	0	0	12	176	380
	HDD65	4300	951	764	556	291	49	5	0	0	19	229	595	841
	CDD50	4063	2	6	60	205	498	723	886	808	579	262	30	4
	CDD65	1300	0	0	2	11	82	278	422	343	148	14	0	0
	CDH74	15293	0	2	33	196	1040	3214	4835	4078	1622	267	6	0
	CDH80	7006	0	0	6	39	321	1528	2497	1981	579	55	0	0
Wind	WSAvg	4.8	3.3	4.1	5.3	5.2	5.5	6.7	7.2	6.0	5.0	4.0	2.8	3.0
Precipitation	PrecAvg	11.9	1.5	1.5	2.3	1.8	1.2	0.5	0.2	0.1	0.2	0.4	1.1	1.2
	PrecMax	17.5	2.5	2.8	4.5	3.7	2.6	1.4	1.1	0.8	1.2	1.4	2.6	2.2
	PrecMin	7.0	0.6	0.6	0.7	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.3
	PrecStd	2.2	0.6	0.6	1.0	0.9	0.6	0.4	0.3	0.2	0.3	0.4	0.7	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	61.2	66.5	77.7	84.6	91.2	98.5	102.0	99.8	93.7	86.0	73.8	63.0
		MCWB	44.0	49.7	53.2	59.2	61.5	63.1	65.7	64.2	61.5	58.6	53.3	45.1
	2%	DB	56.8	60.5	71.6	79.2	86.7	94.8	98.2	95.0	89.7	80.5	66.7	57.4
		MCWB	42.8	45.9	51.3	57.4	60.8	62.7	64.9	63.3	61.0	57.4	51.5	44.1
	5%	DB	51.8	55.4	66.5	75.1	83.7	91.6	94.8	92.3	86.2	76.7	62.5	53.7
		MCWB	41.3	43.6	49.6	55.7	60.0	61.7	65.0	62.3	61.1	56.5	49.6	42.6
	10%	DB	46.9	51.1	61.8	70.2	80.3	88.7	91.6	89.7	82.8	72.5	58.7	49.7
		MCWB	38.8	42.1	48.4	54.2	59.6	61.6	64.7	62.2	60.4	55.4	48.0	41.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.6	50.5	55.7	62.1	66.0	69.7	75.7	72.1	69.3	62.4	55.2	48.4
		MCDB	56.4	63.1	71.2	76.3	81.3	86.9	90.6	86.7	82.8	77.4	67.8	59.3
	2%	WB	43.8	47.3	53.1	59.4	63.8	67.4	71.6	70.2	66.3	60.2	53.0	45.9
		MCDB	54.3	57.7	67.3	74.9	79.8	83.4	86.4	83.9	79.3	73.7	64.5	53.8
	5%	WB	41.7	45.0	51.1	57.3	62.4	66.1	70.0	68.2	64.7	58.6	50.9	43.8
		MCDB	50.5	53.9	64.0	71.1	77.5	82.4	85.2	82.1	77.6	71.4	59.8	51.1
	10%	WB	39.4	42.6	49.0	55.4	60.9	64.8	68.7	66.5	63.2	57.0	48.8	41.8
		MCDB	46.0	50.1	60.1	67.4	75.3	80.7	83.3	80.6	76.1	68.6	56.9	48.2
Mean Daily Temperature Range	5% DB	MDBR	18.9	19.4	21.4	22.4	23.7	25.2	24.7	26.8	26.3	25.3	21.5	18.6
		MCDBR	24.6	26.6	29.6	30.2	31.0	31.1	30.3	31.2	32.0	32.0	29.0	24.8
		MCWBR	14.3	15.1	13.7	13.2	11.9	10.9	10.8	12.5	14.0	15.2	16.5	14.8
	5% WB	MCDBR	22.4	24.0	27.1	26.6	26.2	24.3	23.9	24.2	24.6	26.8	24.5	21.3
		MCWBR	13.7	14.2	14.0	12.7	11.7	9.6	9.4	10.4	11.5	13.2	14.5	13.8
Clear-Sky Solar Irradiance	taub	0.330	0.372	0.450	0.503	0.515	0.497	0.507	0.454	0.418	0.424	0.367	0.327	
	taud	2.257	2.123	1.919	1.842	1.849	1.905	1.883	2.011	2.086	2.035	2.174	2.278	
	Ebn at Noon	266	267	254	249	248	252	248	259	262	244	247	258	
	Edh at Noon	32	42	56	64	65	62	62	54	47	45	35	29	
All-Sky Solar Radiation	RadAvg	719	962	1332	1736	2111	2487	2441	2284	1871	1326	884	655	
	RadStd	64	95	103	170	106	71	67	104	63	66	61	76	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)	N/A	N/A	N/A	+0.94	-1.03	-1.15	N/A	N/A	N/A	+148

Nomenclature: See separate page